

Manuale tecnico

Requisiti

- PHP versione 8.2
- PostgreSQL 16.6

Si assume che il progetto venga eseguito su una distribuzione Linux, e' stato testato su NixOS 24.11; il file default.nix serve in questo sistema operativo per evitare installazioni globali.

Schema concettuale (ER)

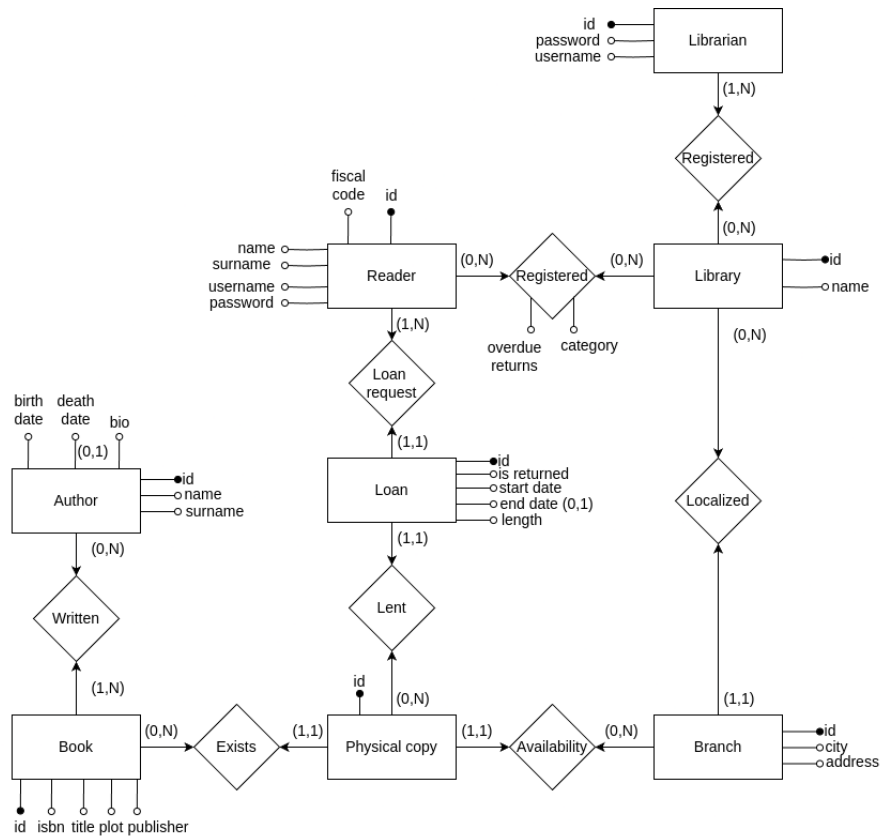


Figure 1: er

Schema logico (relazionale)

```
author(<ins>id</ins>, name, surname, bio,  
      birth_date, death_date*)
```

```

book(<ins>id</ins>, isbn, title, publisher, plot)
author_book(<ins>id_author, id_book</ins>)
physical_copy(<ins>id</ins>, id_book, id_branch)
loan(<ins>id</ins>, id_reader, id_physical_copy,
    start_date, end_date*, length, is_returned)
reader(<ins>id</ins>, fiscal_code, username,
    password, name, surname)
librarian(<ins>id</ins>, username, password, email)
library(<ins>id</ins>, name)
library_reader(<ins>id_library, id_reader</ins>,
    overdue_returns, category)
library_librarian(<ins>id_library, id_librarian</ins>)
branch(<ins>id</ins>, id_library, city, address)

```

Configurazione

Il progetto non ha bisogno di credenziali di superuser per poter essere eseguito. I seguenti comandi si intendono come eseguiti dalla root del progetto.

Creare un database seguendo i dati di configurazione nella cartella scripts, eseguire il setup iniziale e far partire il servizio di PostgreSQL

```

$ ./scripts/postgresql-setup.sh
$ ./scripts/postgresql-start.sh

```

Far partire PHP in ascolto localmente

```

$ ./scripts/php-start.sh

```

Si può fare reset del database per riportarsi in una situazione iniziale pre-configurata

```

$ ./scripts/postgresql-reset.sh

```

Per fermare il servizio di PostgreSQL

```

$ ./scripts/postgresql-stop.sh

```

Funzionalità realizzate con strutture interne della base di dati

Blocco prestiti a lettori ritardatari

Requisito: un prestito può essere concesso solo se il lettore che lo richiede ha meno di 5 riconsegne in ritardo all'attivo.

Implementazione: la funzione `check_overdue_loans()` e il relativo trigger `trigger_check_overdue_loans` impediscono l'inserimento di un nuovo prestito se il lettore ha già 5 o più riconsegne in ritardo. La verifica viene effettuata controllando il numero di ritardi registrati nella tabella `library_reader`.

Numero massimo di prestiti

Requisito: I lettori di categoria “base” possono avere al massimo 3 volumi in prestito, mentre i lettori di categoria “premium” possono averne al massimo 5.

Implementazione: la funzione `ensure_max_loans()` e il relativo trigger `trigger_ensure_max_loans` verificano il numero di prestiti attivi per un lettore in base alla sua categoria. Se il numero massimo consentito è raggiunto, il sistema genera un’eccezione e impedisce il nuovo prestito.

Ritardi nelle restituzioni

Requisito: alla restituzione di un volume, se effettuata in ritardo, è necessario aggiornare il contatore dei ritardi del lettore.

Implementazione: la funzione `update_overdue_returns()` aggiorna il contatore `overdue_returns` nella tabella `library_reader` ogni volta che un prestito viene contrassegnato come restituito (`is_returned = TRUE`) e la restituzione è avvenuta dopo la data di scadenza.

Disponibilità dei volumi

Requisito: un prestito può essere concesso solo se il volume richiesto è disponibile.

Implementazione: la funzione `check_physical_copy_availability()` e il relativo trigger `prevent_duplicate_loans` impediscono l’inserimento di un nuovo prestito se la copia fisica è già in prestito (ovvero, esiste un prestito attivo per la stessa copia).

Proroga della durata di un prestito

Requisito: la proroga della durata di un prestito può essere concessa solo se il prestito non è già in ritardo.

Implementazione: la funzione `extend_loan()` consente di estendere la durata di un prestito solo se la data di scadenza originale è successiva o uguale alla data corrente, impedendo così la proroga di prestiti già scaduti.

Selezione della sede

Requisito: se il lettore specifica una sede per il prestito, il prestito può avvenire solo su copie presenti nella sede specificata. Se non disponibili, possono essere considerate copie di altre sedi previo avviso al lettore.

Implementazione: questo requisito viene gestito dall’applicazione che interagisce con la base di dati. Tuttavia, la struttura della tabella `physical_copy` permette di verificare la disponibilità di copie presso una specifica sede prima di procedere al prestito.

Statistiche per ogni sede

Requisito: per ogni sede, è necessario mantenere il numero totale delle copie gestite, il numero totale dei codici ISBN gestiti e il numero totale di prestiti attivi.

Implementazione: la vista materializzata `branch_stats` fornisce una panoramica aggiornata sulla disponibilità e l'attività di prestito di ciascuna sede, aggregando i dati relativi a copie fisiche e prestiti in corso.

Ritardi per ogni sede

Requisito: generare un report per ogni sede con i libri in prestito in ritardo e i lettori che li hanno in carico.

Implementazione: la vista `overdue_readers_view` raccoglie informazioni sui lettori con prestiti in ritardo, collegandoli alla sede della biblioteca a cui appartengono. Questo consente di generare report dettagliati sui ritardi per ogni sede.

Struttura del progetto

Segue una spiegazione di come e' strutturato il progetto

config

Contiene la configurazione di PostgreSQL, su NixOS solo alcune cartelle sono in scrittura (ad esempio la home dell'utente) quindi ho preferito far partire l'istanza del database nella cartella corrente, per fare questo ho dovuto modificare la posizione della socket

db

Contiene il dump del database, da cui si puo' ripartire agevolmente con l'apposito script nel caso si facciano modifiche indesiderate in fase di esposizione.

docs

Contiene la documentazione del progetto.

pgsql

Contiene i dati relativi all'istanza di PostgreSQL.

scripts

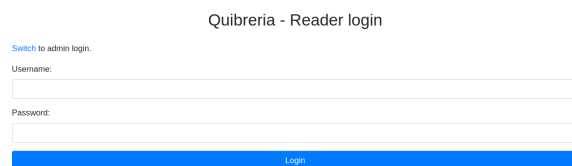
Contiene gli script utili a far partire il progetto.

src

Suddivisa in admin/ e reader/ in base al tipo di utente che ha effettuato il login.
Eventuali file condivisi sono dentro src/ stessa.

Screenshot di funzionamento

Lettore



Quibreria - Reader login

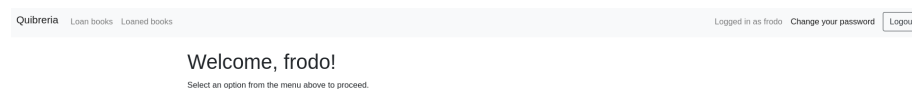
[Switch to admin login.](#)

Username:

Password:

Login

Figure 2: Login



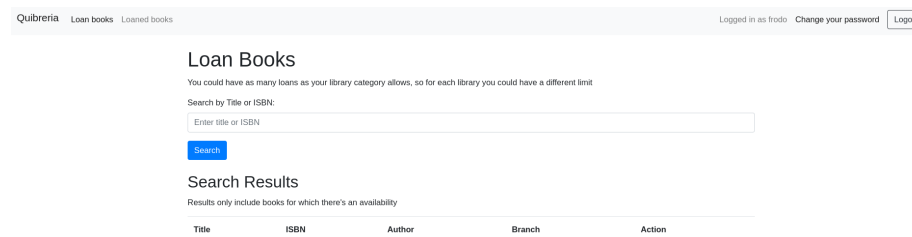
Quibreria Loan books Loaned books

Logged in as frodo Change your password Logout

Welcome, frodo!

Select an option from the menu above to proceed.

Figure 3: Index



Quibreria Loan books Loaned books

Logged in as frodo Change your password Logout

Loan Books

You could have as many loans as your library category allows, so for each library you could have a different limit

Search by Title or ISBN:

Enter title or ISBN

Search

Search Results

Results only include books for which there's an availability

Title	ISBN	Author	Branch	Action
-------	------	--------	--------	--------

Figure 4: Loan books

Bibliotecario

Quibreria

Loan books

Loaned books

Logged in as frodo

Change your password

Logout

Loan Books

You could have as many loans as your library category allows, so for each library you could have a different limit

Search by Title or ISBN:

the

Search

Search Results

Results only include books for which there's an availability

Title (ISBN)	Author	Library	Branch	Action
The Old Man and the Sea (978-0684801223)	Ernest Hemingway	Minas Tirith Library	Minas Tirith - 789 White Tree Road	Loan request
The Lord of the Rings (978-0618640157)	John Tolkien	Dorian Gray Library	Oxford - 987 Piccadilly Square	Loan request
The Pillars of the Earth (978-0451225245)	Ken Follett	Jedi Archive	Alderaan - 222 Peaceful Grove	Loan request

Figure 5: Loan books dopo una ricerca per titolo

Quibreria

Loan books

Loaned books

Logged in as frodo

Change your password

Logout

Book loan requested successfully

Loan Books

You could have as many loans as your library category allows, so for each library you could have a different limit

Search by Title or ISBN:

Enter title or ISBN

Search

Search Results

Results only include books for which there's an availability

Title	ISBN	Author	Branch	Action
-------	------	--------	--------	--------

Figure 6: Loaned book messaggio

Quibreria

Loan books

Loaned books

Logged in as frodo

Change your password

Logout

Loaned Books

Listed here are books that you loaned, to return a book bring it to its branch

You currently have no active loans.

Figure 7: Niente prestiti per questo utente

Quibreria

Loan books

Loaned books

Logged in as frodo

Change your password

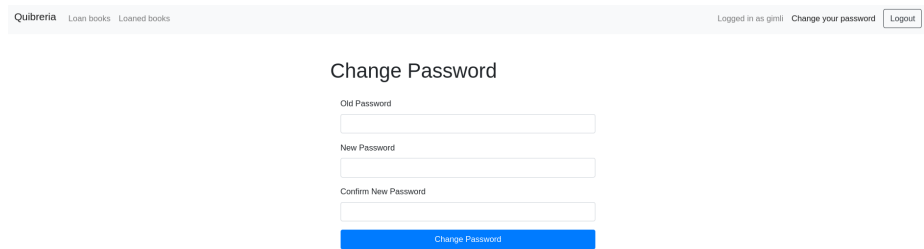
Logout

Loaned Books

Listed here are books that you loaned, to return a book bring it to its branch

Title	ISBN	Branch (City)	Branch (Address)	Start date	Loan length (days)
The Lord of the Rings	978-0618640157	Portland	123 Oak Street	2025-02-15	30
The Old Man and the Sea	978-0684801223	Portland	123 Oak Street	2025-02-15	30
The Pillars of the Earth	978-0451225245	Oxford	987 Piccadilly Square	2025-02-15	30

Figure 8: Lettore con un eccellente gusto in fatto di libri



Quibreria Loan books Loaned books

Logged in as gimli [Change your password](#) [Logout](#)

Change Password

Old Password

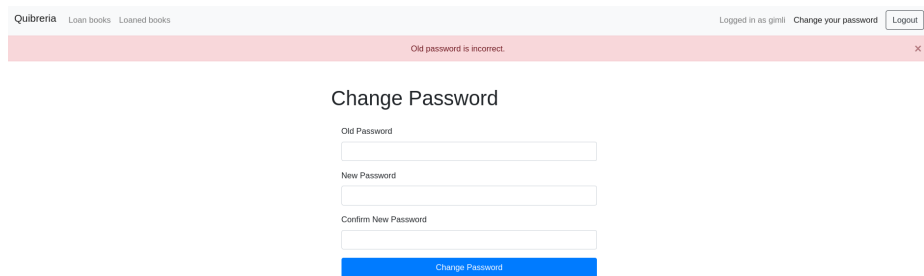
New Password

Confirm New Password

[Change Password](#)

This figure shows a web form for changing a password. The form is titled "Change Password" and is located on a page with a header containing the site name "Quibreria" and navigation links "Loan books" and "Loaned books". The header also shows the user is logged in as "gimli" and provides links to "Change your password" and "Logout". The form itself consists of three input fields: "Old Password", "New Password", and "Confirm New Password", each with a password icon. Below the fields is a blue button labeled "Change Password". A green success message "Password changed" is displayed at the bottom of the form area.

Figure 9: Change password



Quibreria Loan books Loaned books

Logged in as gimli [Change your password](#) [Logout](#)

Old password is incorrect. ✕

Change Password

Old Password

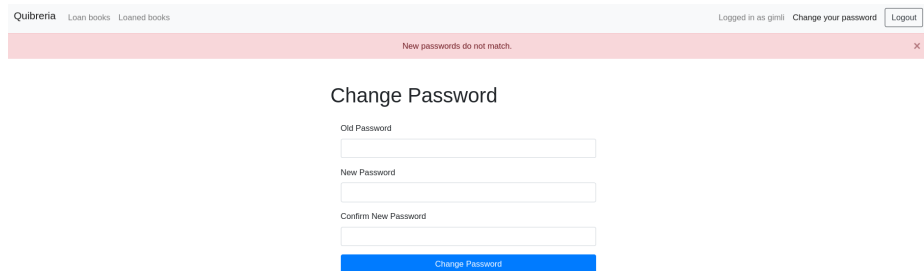
New Password

Confirm New Password

[Change Password](#)

This figure shows the same "Change Password" form as Figure 9, but with an error message. A red banner at the top of the form area displays the text "Old password is incorrect." followed by a red "X" icon. The form fields and the "Change Password" button remain visible below the error message.

Figure 10: Change password fallita perche' la vecchia password non e' corretta



Quibreria Loan books Loaned books

Logged in as gimli [Change your password](#) [Logout](#)

New passwords do not match. ✕

Change Password

Old Password

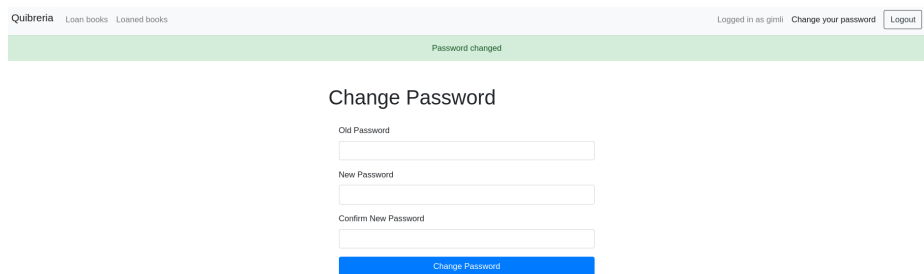
New Password

Confirm New Password

[Change Password](#)

This figure shows the "Change Password" form with a different error message. A red banner at the top of the form area displays the text "New passwords do not match." followed by a red "X" icon. The form fields and the "Change Password" button are still visible below the error message.

Figure 11: Change password fallita perche' le password non corrispondono



Quibreria Loan books Loaned books

Logged in as gimli [Change your password](#) [Logout](#)

Password changed

Change Password

Old Password

New Password

Confirm New Password

[Change Password](#)

This figure shows the "Change Password" form with a success message. A green banner at the top of the form area displays the text "Password changed". The form fields and the "Change Password" button are visible below the success message.

Figure 12: Change password successo

Quibreria - Librarian login

[Switch to reader login.](#)

Username:

Password:

Login

Figure 13: Login

Quibreria Manage Readers Manage Books Manage Branches Manage Loans

Logged in as conan Change your password Logout

Welcome, Conan The Librarian!

Select an option from the menu above to proceed.

Figure 14: Index

Quibreria Manage Readers Manage Books Manage Branches Manage Loans

Logged in as conan Change your password Logout

Manage Readers

Username:

Password:

Fiscal Code:

Name:

Surname:

Add Reader

Search by username, name, or surname Search

Readers List

Listed here are readers that are registered to your libraries

Username	Library name	Overdue count	Fiscal Code	Name	Surname	Action
No readers found.						

Figure 15: Manage readers

Quibreria
Manage Readers
Manage Books
Manage Branches
Manage Loans

Logged in as conan
Change your password
Logout

Manage Readers

Username:

Password:

Fiscal Code:

Name:

Surname:

Add Reader

Search

Readers List

Listed here are readers that are registered to your libraries

Username	Library name	Overdue count	Fiscal Code	Name	Surname	Action
frodo	Shewshank Library	0	FRDBG1234567891	Frodo	Baggins	Reset overdue Edit Delete

Figure 16: Manage readers, trovato un reader attraverso la ricerca

Quibreria
Manage Readers
Manage Books
Manage Branches
Manage Loans

Logged in as conan
Change your password
Logout

Manage Readers

Username:

Password:

Fiscal Code:

Name:

Surname:

Update Reader

Search by username, name, or surname

Search

Readers List

Listed here are readers that are registered to your libraries

Username	Library name	Overdue count	Fiscal Code	Name	Surname	Action
No readers found.						

Figure 17: Manage readers, modifica

Quibreria

Manage Readers

Manage Books

Manage Branches

Manage Loans

Logged in as [admin](#)

[Change your password](#)

Logout

Manage books

ISBN:

Title:

Publisher:

Plot:

Authors:

John Tolkien

Ernest Hemingway

Richard Bach

Ken Follett

Branch:

Portland 123 Oak Street

Add Book

Search by ISBN or title

Search

Books List

Listed here are books that are available in your libraries

ISBN	Title	Publisher	Plot	Branch	Action
------	-------	-----------	------	--------	--------

Figure 18: Manage books

The

Search

Books List

Listed here are books that are available in your libraries

ISBN	Title	Publisher	Plot	Branch	Action
978-0618640157	The Lord of the Rings	George Allen & Unwin	An epic high-fantasy novel written by English author and scholar J. R. R. Tolkien. The story began as a sequel to Tolkien's 1937 fantasy novel The Hobbit, but eventually developed into a much larger work.		Edit Delete
978-0684801223	The Old Man and the Sea	Charles Scribner's Sons	A short novel written by the American author Ernest Hemingway in 1951 in Cuba, and published in 1952. It tells the story of Santiago, an aging Cuban fisherman who struggles with a giant marlin far out in the Gulf Stream.		Edit Delete
978-0451225245	The Pillars of the Earth	William Morrow & Company	A historical novel by Ken Follett, published in 1989, about the building of a cathedral in the fictional town of Kingsbridge, England. It is the first in the Kingsbridge Series.		Edit Delete

Figure 19: Manage books, trovati libri che iniziano per “The”

Quibreria

Manage Readers

Manage Books

Manage Branches

Manage Loans

Logged in as conan

Change your password

Logout

Manage Books

ISBN:

978-0618640157

Title:

The Lord of the Rings

Publisher:

George Allen & Unwin

Plot:

An epic high-fantasy novel written by English author and scholar J. R. R. Tolkien. The story began as a sequel to Tolkien's 1937 fantasy novel The Hobbit, but eventually developed into a much larger work.

Authors:

John Tolkien

Ernest Hemingway

Richard Bach

Ken Follett

Branch:

Portland 123 Oak Street

Update Book

Search by ISBN or title

Search

Books List

Listed here are books that are available in your libraries

ISBN	Title	Publisher	Plot	Branch	Action
------	-------	-----------	------	--------	--------

Figure 20: Manage books, edit

Quibreria

Manage Readers

Manage Books

Manage Branches

Manage Loans

Logged in as conan

Change your password

Logout

Manage Branches

Library:

Shawshank Library

City:

Address:

Add Branch

Search by city, address, or library

Search

Branches List

Listed here are branches that belong to your libraries

Library	City	Address	Total copies	Total distinct ISBNs	Active loans	Action
No branches found						

Figure 21: Manage branches

Quibreria

Manage Readers

Manage Books

Manage Branches

Manage Loans

Logged in as conan

Change your password

Logout

Manage Branches

Library:

Shawshank Library

City:

Address:

Add Branch

Rivendell

Search

Branches List

Listed here are branches that belong to your libraries

Library	City	Address	Action
Rivendell Library	Rivendell	333 Elven Path	Statistics List returns report Edit Delete
Rivendell Library	Lothlórien	444 Golden Forest	Statistics List returns report Edit Delete

Figure 22: Manage branches, trovate le sedi di Rivendell

Quibreria

Manage Readers

Manage Books

Manage Branches

Manage Loans

Logged in as conan

Change your password

Logout

Manage Branches

Library:

Shawshank Library

City:

Address:

Add Branch

Search by city, address, or library

Search

Branches List With Stats

Listed here are branches stats that belong to your libraries

Library	City	Address	Total copies	Total distinct ISBNs	Active loans
Rivendell Library	Rivendell	333 Elven Path	10	4	0

Figure 23: Manage branches, statistiche per sede di Rivendell

Quibreria

Manage Readers

Manage Books

Manage Branches

Manage Loans

Logged in as conan

Change your password

Logout

Manage Branches

Library:

Rivendell Library

City:

Rivendell

Address:

333 Elven Path

Update Branch

Search by city, address, or library

Search

Branches List

Listed here are branches that belong to your libraries

Library	City	Address	Total copies	Total distinct ISBNs	Active loans	Action
No branches found						

Figure 24: Manage branches, edit

Quibreria
Manage Readers
Manage Books
Manage Branches
Manage Loans

Logged in as conan
Change your password
Logout

Manage Loans

Listed here are loans that are not yet returned to your libraries

Book	Reader	Loan date	Days left	Action
Jonathan Livingston Seagull	Ginli	2020-03-01	-1798	Mark as Returned Extend Loan

Figure 25: Manage loans

Quibreria
Manage Readers
Manage Books
Manage Branches
Manage Loans

Logged in as conan
Change your password
Logout

Loan is already overdue and cannot be extended

Manage Loans

Listed here are loans that are not yet returned to your libraries

Book	Reader	Loan date	Days left	Action
Jonathan Livingston Seagull	Ginli	2020-03-01	-1798	Mark as Returned Extend Loan

Figure 26: Manage loans, non si puo’ estendere un loan overdue

Quibreria
Manage Readers
Manage Books
Manage Branches
Manage Loans

Logged in as conan
Change your password
Logout

Change Password

Old Password

New Password

Confirm New Password

Change Password

Figure 27: Change password, come per il lettore